another individual's life, in particular a child's life. A measure of energy such as the calorie is tempting, and some ecologists have devoted themselves to the accounting of energy costs in nature. This is inadequate though, because it is only loosely convertible into the currency that really matters, the 'gold-standard' of evolution, gene survival. R. L. Trivers, in 1972, neatly solved the problem with his concept of *parental investment* (although, reading between the close-packed lines, one feels that Sir Ronald Fisher, the greatest biologist of the twentieth century, meant much the same thing in 1930 by his 'parental expenditure').*

Parental investment (P.I.) is defined as 'any investment by the parent in an individual offspring that increases the offspring's chance of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring'. The beauty of Trivers's parental investment is that it is measured in units very close to the units that really matter. When a child uses up some of its mother's milk, the amount of milk consumed is measured not in pints, not in calories, but in units of detriment to other children of the same mother. For instance, if a mother has two babies, *X* and *Y*, and *X* drinks one pint of milk, a major part of the P.I. that this pint represents is measured in units of increased probability that *Y* will die because he did not drink that pint. P.I. is measured in units of decrease in life expectancy of other children, born or yet to be born.

Parental investment is not quite an ideal measure, because it overemphasizes the importance of parentage, as against other genetic relationships. Ideally we should use a generalized *altruism investment* measure. Individual *A* may be said to invest in individual *B*, when *A* increases *B*'s chance of surviving, at the cost of *A*'s ability to invest in other individuals including herself, all costs being weighted by the appropriate relatedness. Thus a parent's investment in any one child should ideally be measured in terms of detriment to life expectancy not only of other children, but also of nephews, nieces, herself, etc. In many respects, however, this is just a quibble, and Trivers's measure is well worth using in practice.

Now any particular adult individual has, in her whole lifetime, a certain total quantity of P.I. available to invest in children (and other relatives and in herself, but for simplicity we consider only children). This represents the sum of all the food she can gather or manufacture in a